

CS 450: Assignment 03

Programming Assignments (95%)

- **This assignment is in C++ and GLSL.**
- Follow the exercise instructions from the following slide decks:
 - o **08 "Vertex and Index Buffers"**
- **Required new header/library files (as written in exercises):**
 - o `src/include/student/`
 - `VKBuffer.hpp`
 - `VKMesh.hpp`
 - o `src/lib/student/`
 - `VKBuffer.cpp`
 - `VKMesh.cpp`
- **Main application setup:**
 - o Copy `src/app/Exercises08.cpp` and name it `src/app/Assign03.cpp`
 - o Change the **application name** from "Exercises08" to "Assign03"
 - o Change the **window title** to "Assign03: " + your SITNET ID
 - E.g., "Assign03: realemj"
- **Main application modification:**
 - o In `Assign03.cpp`, create something interesting in terms of geometry with the following restrictions:
 - **Winding order MUST be COUNTER-CLOCKWISE.**
 - No vertex should be outside of the window (no x or y value greater than 1.0 or less than -1.0).
 - All vertex z coordinates should be set to 0.5.
 - Triangles should NOT overlap each other.
 - You should have at least **5 vertices** and **3 triangles**.
- **Shader code setup:**
 - o Copy folder `vulkanshaders/Exercises08` as `vulkanshaders/Assign03`
- **CMake setup:**
 - o Modify `CMakeLists.txt` by adding the following line to the end of the file:
 - `CREATE_VULKAN_EXECUTABLE(Assign03)`

Screenshot (5%)

- **Take a screenshot** of the **APPLICATION** window.
- Copy the image into the **screenshots** folder as "Assign03.png"